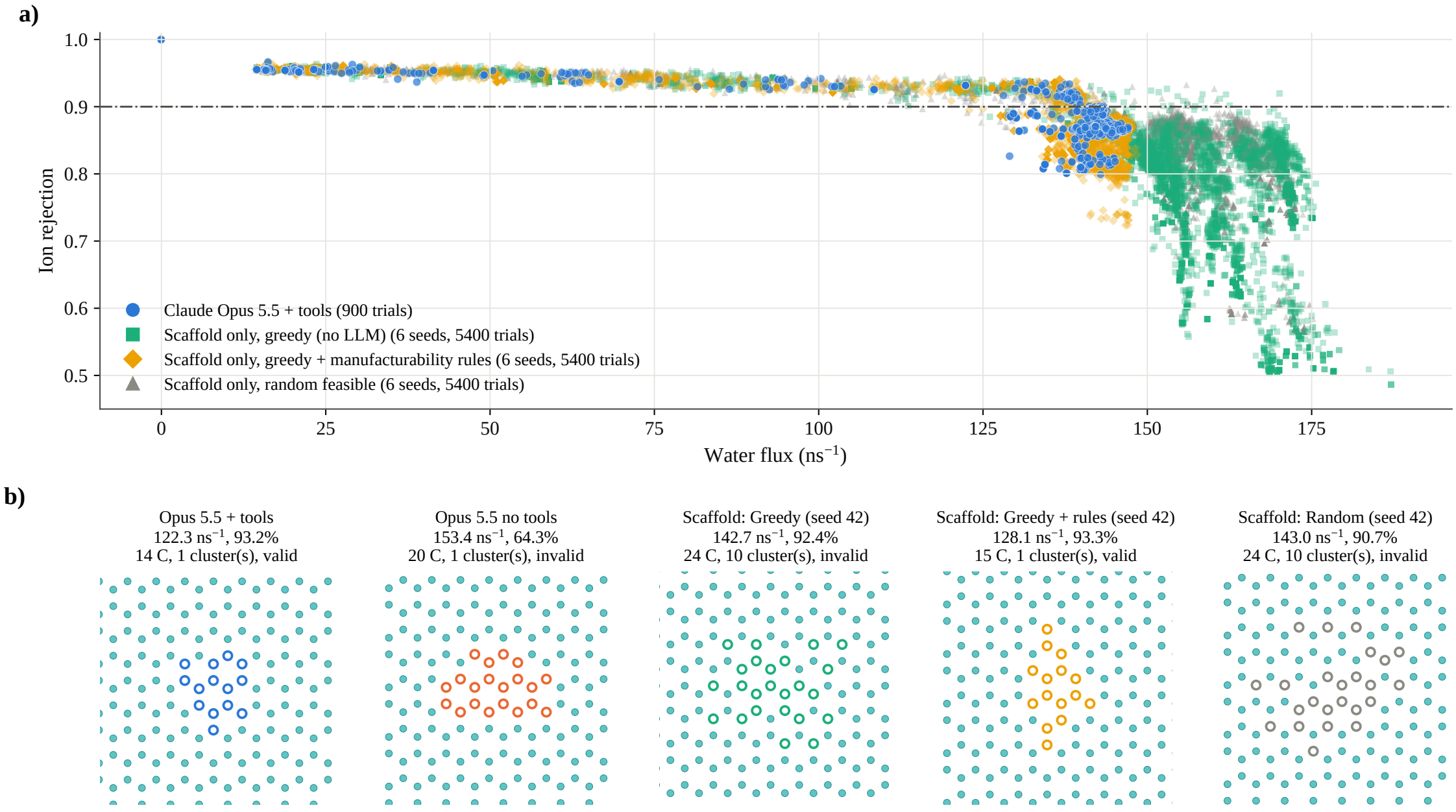

a)
1.0
0.9
0.8
0.7
0.6
0.5
Ion rejection
0
25
50
75
100
125
150
175
Water flux (ns^{-1})
Claude Opus 5.5 + tools (900 trials)
Scaffold only, greedy (no LLM) (6 seeds, 5400 trials)
Scaffold only, greedy + manufacturability rules (6 seeds, 5400 trials)
Scaffold only, random feasible (6 seeds, 5400 trials)
b)
Opus 5.5 + tools
122.3 ns^{-1}, 93.2%
14 C, 1 cluster(s), valid
Opus 5.5 no tools
153.4 ns^{-1}, 64.3%
20 C, 1 cluster(s), invalid
Scaffold: Greedy (seed 42)
142.7 ns^{-1}, 92.4%
24 C, 10 cluster(s), invalid
Scaffold: Greedy + rules (seed 42)
128.1 ns^{-1}, 93.3%
15 C, 1 cluster(s), valid
Scaffold: Random (seed 42)
143.0 ns^{-1}, 90.7%
24 C, 10 cluster(s), invalid